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WaveDesignerDefTopic

	
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About WaveDesigner

About WaveDesigner

WaveDesigner allows you to create simple or compound waveform definitions. Using a predefined list of waveform primitives, such as sine and ramp, you can specify waveshape parameters and create reusable waveform definitions to be applied to an instrument.

The WaveDesigner offers the following features and capabilities:

- | |
|---|
| <ul style="list-style-type: none">• A set of waveform primitives including: sine, ramp, staircase, pulse, DC, multitoneLab, ExcelRange, and VBT |
| <ul style="list-style-type: none">• Ability to create simple or compound wave definitions independent of timing and instrument specific information |

- [Ability to save waveform definitions to a Wave Definitions sheet to be accessed at another time](#)
- [Integration with WaveScope to preview waveforms you create](#)

This section includes the following topics:

- [Building Waveforms](#)
- [Waveform Definitions](#)
- [Waveform Primitives](#)

The WaveDesigner information includes the following sections:

- [WaveDesigner Display](#)
- [Creating Wave Definitions](#)

For more information, see:

- [WaveDefinitions](#) in the VBT Syntax Reference
- [MSW and the WaveDesigner](#)
- [Wave Definitions Sheet](#)
- [About WaveScope](#)

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About WaveDesigner : Building Waveforms

Building Waveforms

In IG-XL, creating a waveform requires:

- Creation of wave definitions using the WaveDesigner tool
- Creation of a mixed signal timing set using the Mixed Signal Workshop (MSW) or VBT language

WaveDesigner is the graphical interface tool you use to create simple or compound wave definitions which define a waveform's shape. Wave definitions contain only waveshape information and are independent of amplitude, timing, and instrument specific information so that they can be used on any instrument with any timing.

You add timing information to a wave definition in a separate process using the Mixed Signal Workshop or VBT language. See Mixed Signal Workshop.

WaveDesigner provides a number of basic waveform primitives, such as sine and ramp, from which you can build waveform definitions. (See Waveform Primitives.)

You can use these primitives to build two types of wave definitions:

- Simple waveform definitions involving only one primitive
- Compound waveform definitions which concatenate primitives and nest wave definitions within the new wave definition

Once you create a wave definition, you can save it to a Waveform Definitions sheet, and it is reusable throughout the test program because it is instrument-independent.

WaveDesigner provides access to an integrated WaveScope tool, so that you can preview the resulting waveform as you design a wave definition.

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About WaveDesigner : Waveform Definitions

Waveform Definitions

A waveform definition is an abstract representation of a waveform that contains no timing, amplitude, or instrument-specific information, only the waveform's basic shape. You add timing information to a wave definition in a separate process using the Mixed Signal Workshop or VBT language. (See [Mixed Signal Workshop](#).)

The WaveDesigner tool helps you to create wave definitions by using primitive waveshapes. The structure of each primitive has associated with it a repeat count, relative frequency, relative amplitude, and relative offset, where "relative" means in relation to other components of the definition.

Using WaveDesigner primitives, you can build two types of wave definitions:

- Simple waveform definitions involving only one or more primitive
- Compound waveform definitions which concatenate primitives and nest other wave definitions within the new wave definition

You can save wave definitions to a Waveform Definitions sheet in the IG-XL workbook. Since wave definitions are independent of timing and instrument restrictions, they are reusable throughout the test program by simply integrating them into different timing contexts in the MSW.

See [Creating Wave Definitions](#) for instructions on creating a wave definition using WaveDesigner's primitives. You may also need to create wave definitions at run time. If so, see [Adding a Wave Definition at Run Time](#) for instructions on using the CreateWaveDefinition method.

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About WaveDesigner : Waveform Primitives

Waveform Primitives

Primitives are the building blocks you use to create a wave definition. WaveDesigner provides a number of primitives which contain only the structural or shape information for the waveform segment. To build a wave definition, select a primitive and specify parameter values and options to generate samples for that type of waveform.

WaveDesigner provides following simple primitive types:

Sine	Generates sinusoidal waveforms of arbitrary starting phase. See Using a Sine Primitive.
Ramp	Creates ascending or descending slope waveforms. See Using a Ramp Primitive.
Staircase	Creates ramps with discrete steps. See Using a Staircase Primitive.
Pulse	Generates high or low pulses. See Using a Pulse Primitive.
DC	Creates a constant DC voltage level tone. See Using a DC Primitive.
MultitoneLab	Generates multitone waveforms. See Using a MultitoneLab Primitive.

If these simple primitive types do not generate the samples that you require for your test, two additional primitives are available that allow you to generate samples independent of the basic primitive types.

ExcelRange	Uses a named range in the workbook to create waveform samples. See Using an ExcelRange Primitive.
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VBTCalls a VBT routine that returns an array of samples. See Using a VBT Primitive.

See [Creating Wave Definitions](#) for more information on using primitives to create wave definitions.

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Creating Wave Definitions

Creating Wave Definitions

Typically, you create a wave definition by adding primitives to the Wave Definition table in the WaveDesigner window. The primitive’s component parameters are then assigned to each primitive to build the shape of the waveform. This section describes creating and controlling wave definitions.

The following topics are available:

- | | |
|---|--------------------------------|
| • | Using a DC Primitive |
| • | Using an ExcelRange Primitive |
| • | Using a MultitoneLab Primitive |
| • | Using a Pulse Primitive |
| • | Using a Ramp Primitive |
| • | Using a Sine Primitive |
| • | Using a Staircase Primitive |
| • | Using a VBT Primitive |

- [Adding a Wave Definition at Run Time](#)

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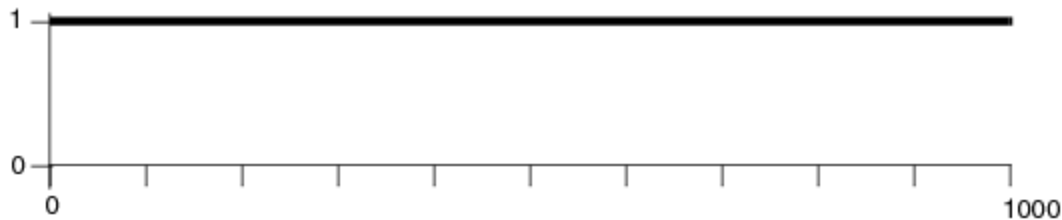
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Creating Wave Definitions : Using a DC Primitive

Using a DC Primitive

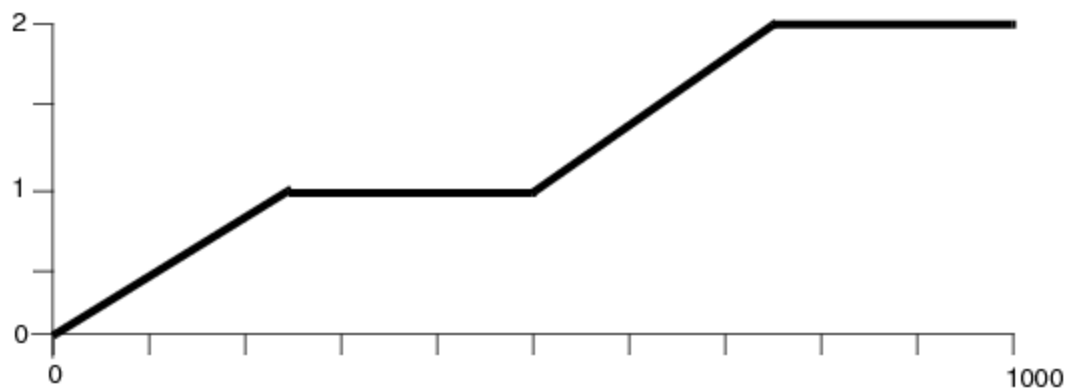
A DC primitive is a constant voltage level that is often used to transition between waveforms resulting in a filler (time delay) between primitives or wave definitions. By default, the DC level is positioned at an amplitude of 1.



DC Primitive

Unlike other primitives, a DC primitive does not have any component parameters. However, you can still control this primitive. By editing the Relative Amplitude and Relative Offset parameters in the Wave Definition table, you can move a DC primitive's position up or down relative to other primitive components in a wave definition.

For example, the figure shows DC primitives being used to maintain a level of 1V over the second quarter of the wave definition and then 2V over the final quarter. The table shows that this was accomplished by setting the first DC primitive's relative amplitude to 1, the second ramp's relative amplitude and relative offset to 1, and the second DC primitive's relative amplitude to 2.



Using a DC Primitive as a Transition Between Two Ramp Primitives

Wave Definition Table Settings for DC Primitive Example

Wave Definition	Repeat Count	Period	Relative Amplitude	Relative Offset
myWave		4		
Ramp	1	1	1	0
DC	1	1	1	0
Ramp	1	1	1	1
DC	1	1	2	0

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Creating Wave Definitions : Using an ExcelRange Primitive

Using an ExcelRange Primitive

If you know the sample values of the signal that you want to include in a wave definition, you can use an ExcelRange primitive to create a reusable primitive where the values you specify will determine the signal's shape.

An ExcelRange primitive allows you to specify the sample values of the desired signal shape in a named cell range (group of cells in a column) in an Excel sheet. By providing unique name for the range, you can call the range of samples into a wave definition rather than specifying individual cell values. The values in the range become the sample points of the primitive.

The following sections describe ExcelRange component parameters and provide instructions for using an ExcelRange primitive:

- [ExcelRange Component Parameters](#)
- [Creating an ExcelRange Primitive](#)

Note:

Since you specify a range of data for an ExcelRange primitive, the number of samples (N) allowable for the primitive is fixed. For example, if the defined range contains data in 20 cells, the number of samples is 20 per cycle. This number must be used with all signals that use this primitive.



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Creating Wave Definitions : Using an ExcelRange Primitive : ExcelRange Component Parameters

ExcelRange Component Parameters

The three component parameters associated with an ExcelRange primitive are: Range name, Bandwidth limited, and Required BW.

Range Name

This parameter identifies the name of range of values that will be used to build the signal's shape. When you add a new ExcelRange primitive to a wave definition, the range name component will be empty. You must type in the name of the range in the parameter exactly as it was defined in the workbook. If you cannot remember the exact name of the range, return to the sheet and look it up in the Name Box menu located above the upper-left corner of the sheet.

If an ExcelRange primitive's range name was already specified, the range name is displayed in the Range Name parameter.

Bandwidth limited

Select this parameter when the signal created by an ExcelRange primitive has a finite bandwidth, when the bandwidth is irrelevant, or when the bandwidth equals the signal's Fr. Examples are when you have register data and the bandwidth is irrelevant, and when you have a sine and the bandwidth is equal to the signal's Fr.

This parameter does not have an immediate effect on an ExcelRange primitive in WaveDesigner.

However, the Bandwidth limited parameter is later used by the Mixed Signal Workshop to determine if there are bandwidth related restrictions on the allowable sample rate that can be used with this primitive. When this parameter is selected, no sample rate restrictions (other than Nyquist) are placed on this primitive when Mixed Signal Workshop instantiates the signal. If the parameter is left unselected (default), the required bandwidth multiplier is used to determine bandwidth related sample rate restrictions. The restriction is that the sample rate must be greater than or equal to the signal Fr multiplied by the required bandwidth multiplier.

Required BW

This parameter does not have an immediate effect on an ExcelRange primitive in WaveDesigner. The Mixed Signal Workshop assumes that the signal’s bandwidth is its repeat rate (FR) multiplied by this constant. For band-unlimited signals, the 10%-90% edge rise time as a percent of the period can be approximated as:

(1) $tr \cong 0.35 / \text{RequiredBandwidthMultiplier}$

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Creating Wave Definitions : Using an ExcelRange Primitive : Creating an ExcelRange Primitive

Creating an ExcelRange Primitive

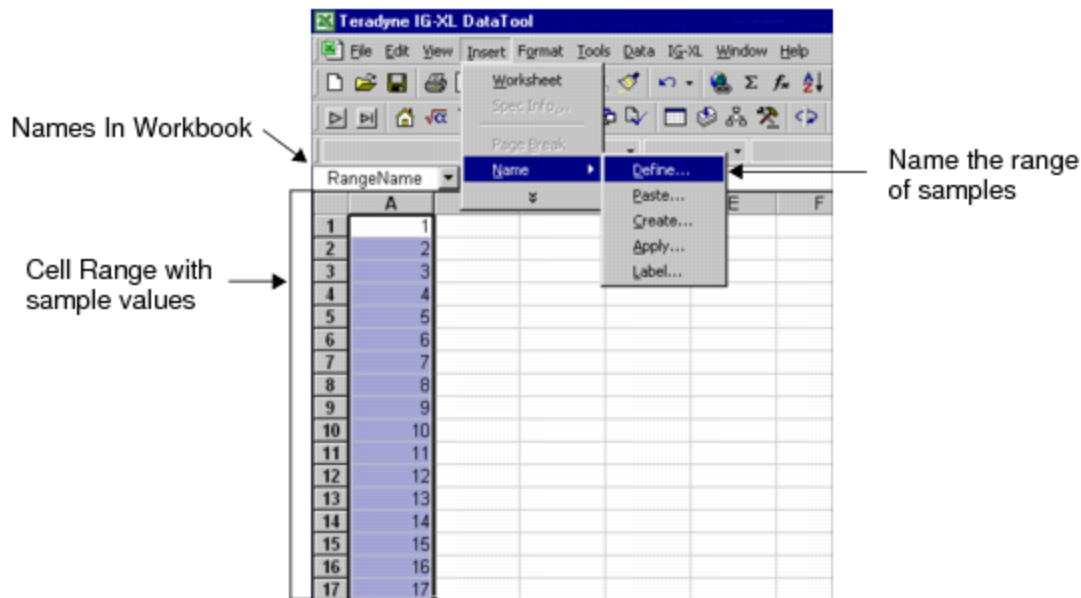
To create an ExcelRange primitive based on user-provided sample values:

1. Create a range of samples in an Excel sheet:
- a. Add a blank Excel sheet to your IG-XL workbook by selecting Insert>IG-XL Worksheet and then selecting Excel Worksheet from the Sheet type menu on the dialog box.

b. Type the sample values of the desired signal in a column. Enter one sample value per cell.

c. Select the range of samples to use on the sheet by holding and dragging the mouse from the first cell in the range to the last cell.

d. Assign a name for the sample range by selecting Insert>Name>Define. Type a name for the range in the Names in workbook field of the Define Name dialog box and click OK.



Creating a Range of Samples in an Excel Sheet

2. Assign the range of samples to an ExcelRange primitive in WaveDesigner:

a. Add a new wave definition or an ExcelRange primitive to the Wave Definition table.

b. Enter the name of the range in the Range name parameter. The name must be exactly as it was defined in the sheet.

3. Display the resulting signal in WaveScope:

a. Enter the actual number of samples that the ExcelRange primitive contains in the Sample Size (N) field on the Preview Scratchpad. The number of samples must match the size of the range.

b. Click the Show Wave button to plot the primitive in the WaveScope display.

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Creating Wave Definitions : Using a MultitoneLab Primitive

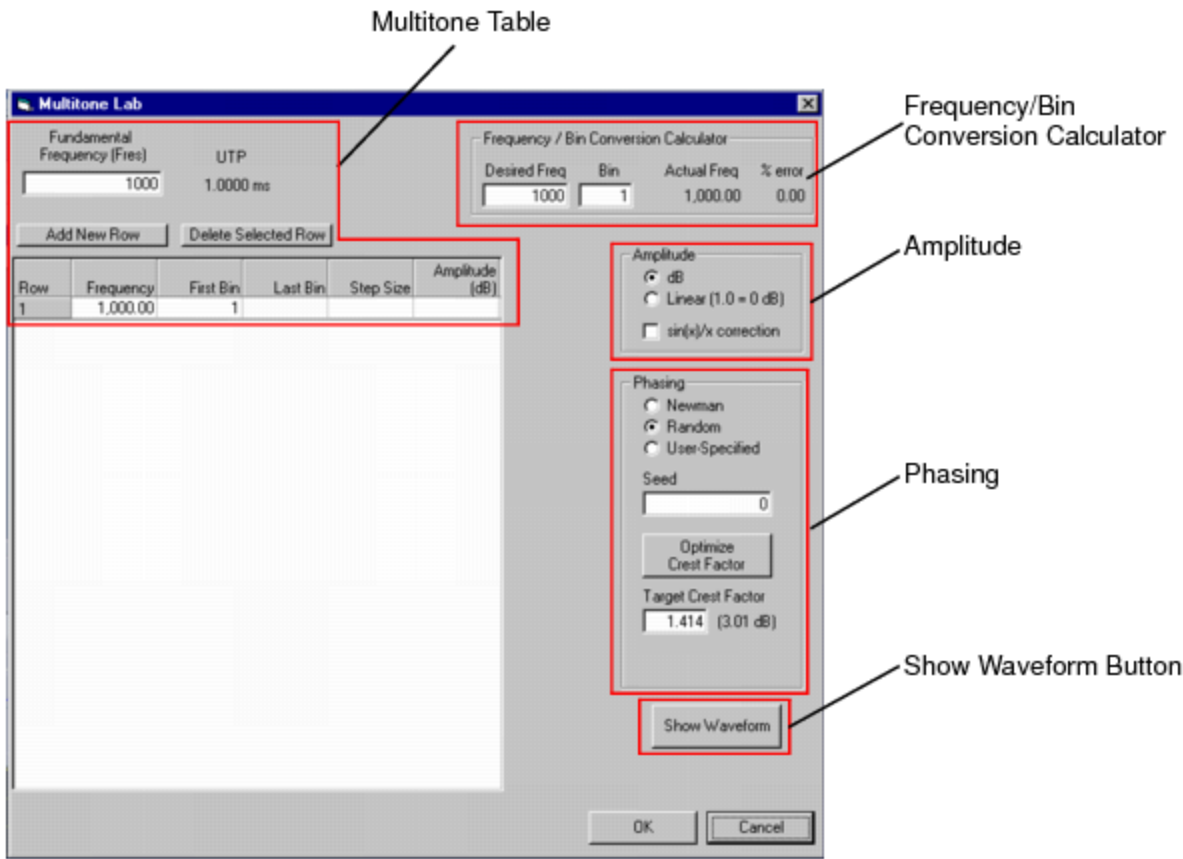
Using a MultitoneLab Primitive

A MultitoneLab primitive allows you to create multitone signals with full control over the tones that compose the signal.

A Multitone primitive does not have typical component parameters like other primitives. It allows you to set its parameters using the MultitoneLab Editor tool.

To access the MultitoneLab Editor:

1. Insert a multitone primitive in a wave definition.
2. Click the MultitoneLab Editor button that appears in the Component Parameter pane. The Multitone Lab Editor window appears.



Multitone Lab Editor

For more information, see:

- [Multitone Lab Editor](#)
- [Creating a Multitone Primitive](#)

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Creating Wave Definitions : Using a MultitoneLab Primitive : Multitone Lab Editor

Multitone Lab Editor

You create a multitone waveform using the Multitone Lab Editor. This topic describes the controls on the display.

The following topics are available:

- Multitone Table
- Frequency/Bin Conversion Calculator
- Amplitude
- Phasing
- Show Waveform Button

For more information, see Using a MultitoneLab Primitive.

Multitone Table

The Multitone table lists the characteristics of each tone in the signal. Each line in the table represents one or more tones. The Multitone Table has the following controls:

Fundamental Frequency (Fres)	The frequency of a bin 1 tone.
---------------------------------	--------------------------------

UTP	Unit Test Period (UTP) is expressed as: $UTP = 1/F_{res}$
Frequency	The frequency of the first tone in this row: $Frequency = First\ Bin \times F_{res}$
First Bin	The bin number of the first tone.
Last Bin	The bin number that the tones cannot exceed. If specified, multiple tones are generated within the limitations of the first and last bins with intervals based on the step size. For example, First Bin, First Bin + Step Size, First Bin+2*Step Size, ... all the way to the Last bin.
Step Size	The size of the interval, or the spacing, between each tone: $Interval\ Size = Step\ Size \times Fundamental\ Frequency$ If the Step Size value is not specified, the step defaults to 1.
Amplitude (dB)	The relative amplitude values of each tone represented on that one line in the table. If left blank, the amplitude defaults to 0 dB.

Frequency/Bin Conversion Calculator

The Frequency/Bin Conversion Calculator is a utility to help translate bins to frequencies, and vice versa. This utility is a read-only calculator, and it does not write results to other parts of the MultitoneLab Editor. If you specify a value in the Desired Freq field, the calculator automatically displays the bin closest to the desired frequency, the actual frequency, and percent error. Similarly, you can enter the bin number and the desired frequency value automatically adjusts.

Desired Freq	The frequency that you want to use in the calculator.
Bin	The bin that this frequency will reside in.
Actual Freq	If the desired frequency cannot be achieved for the specified bin value, the actual frequency of the tone at that bin is displayed.
% error	If the actual frequency is different from the desired frequency, then the percentage of the difference is displayed in this field.

Amplitude

Amplitude controls allow you to select the format unit of the amplitude values for the tones in the signal.

dB	Sets the amplitude value of the tones represented on the last line in the Multitone table to display in decibels (dB). If the Amplitude column in the table is left blank, the amplitude value defaults to 0dB.
Linear (1.0 = 0 dB)	Sets the amplitudes values of the tones represented on the last line in the Multitone table to display in volts. If the Amplitude column in the table is left blank, the amplitude defaults to 1V.
sin(x)/x correction	Applies sin(x)/x correction to the multitone. This option is not recommended for resampled instruments.

Phasing

Phasing controls allow you to find a set of phases for the composite tones to minimize the multitone crest factor. The crest factor is defined as the ratio of the peak to the RMS level of the signal. A large crest factor indicates the signal has rare, but large, deviations from its normal levels. As a rule of thumb, the smaller the crest factor, the easier it is to manipulate, store, and transmit a signal.

Newman	Uses the Newman algorithm to assign starting phases to each tone. This typically generates a low crest factor multitone (typically < 60dB).
Random	Assigns a random starting phase between 0 and 360 degrees to each tone based on the choice of the seed.
User-Specified	Adds a column to the Multitone Table and allows you to specify custom phasing for each tone.
Seed	Determines the starting seed for random phasing as follows: 0 or blank: generates a new seed each time the multitone is created. This results in samples that vary from run to run of the program. - Nonzero: Uses this value as a seed. The phases are initially random, but the samples remain constant from run to run of the program.
Optimize Crest Factor	Click this button to cause the editor to repeatedly generate multitones, remembering the one whose crest factor is closest to the target. The final seed is stored in the Seed field.

When activated, the button changes to a Stop button. Clicking the Stop button halts the generation function and the waveform is automatically plotted in a WaveScope display.

Target Crest Factor

Target for the Optimize Crest Factor. Set to 1.0 to find the smallest possible crest factor.

Show Waveform Button

The Show Waveform launches WaveScope and plots the multitone signal in the display.

		
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Creating Wave Definitions : Using a MultitoneLab Primitive : Creating a Multitone Primitive

Creating a Multitone Primitive

To create a multitone primitive using the MultitoneLab Editor:

1. Specify a value for the Fundamental Frequency.
2. In row 1 of the Multitone Table, set the frequency and bins. If necessary, you can also specify a value for the step size and amplitude.
3. To add more tones to the signal, add additional rows to the Multitone Table. Each row in the table represents one or more tones in the overall signal.
4. Select the phasing.

If you select Random phasing, set the Target Crest Factor and click the Optimize Crest Factor button. This will iterate with different phasings, trying to achieve the crest factors you have selected. If the MultitoneLab Editor can achieve the target crest factor, the tool will stop generating. Otherwise, when you feel the crest factor is close enough to the desired result, click Stop.

5. Click the Show Waveform button to view your multitone signal in a WaveScope display.
6. Close WaveScope. If needed, edit your multitone data.
7. Click OK to save the multitone primitive to WaveDesigner and close the MultitoneLab Editor window.

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Creating Wave Definitions : Using a Pulse Primitive

Using a Pulse Primitive

A Pulse primitive is used to generate high or low pulses. It produces signal shapes in which a signal’s state changes from one or zero, with a short transition period, between the high or low state.

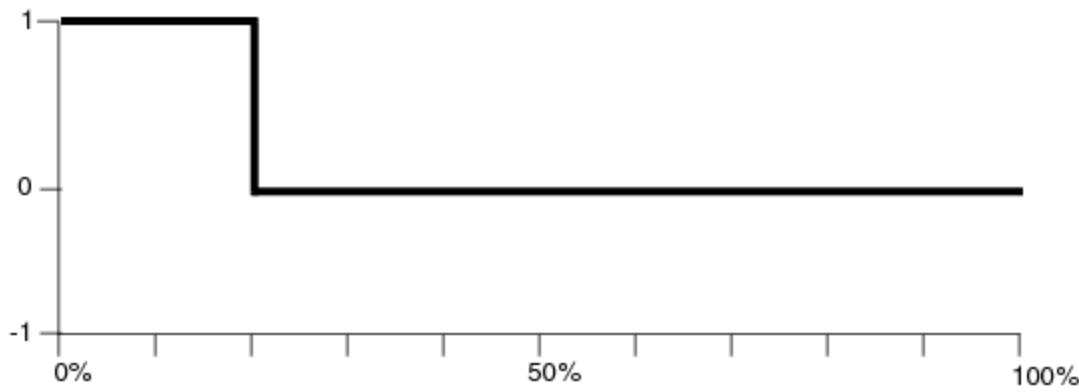
By default, a pulse primitive starts in a high state, unless the Invert pulse? parameter is selected, then it starts in the lower state.

The following parameters allow you to manipulate the positioning of a pulse primitive:

- | | |
|---|--------------------------------|
| • | Duty cycle |
| • | Phase |
| • | Invert pulse? |
| • | Maximum risetime (% of period) |

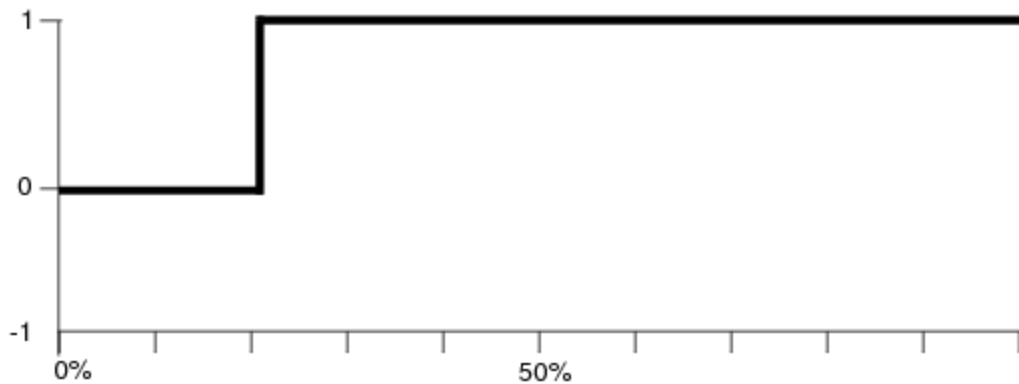
Duty cycle

The duty cycle of a pulse primitive refers to the percent of period for which the first state, either high or low, is constant. A primitive’s duty cycle is expressed as a percentage of the primitive’s entire cycle, where on complete period is 100%. The figure shows that if a pulse primitive’s duty cycle set to 20, the state remains high for 20% of the component’s period and the remaining 80% at the lower level.



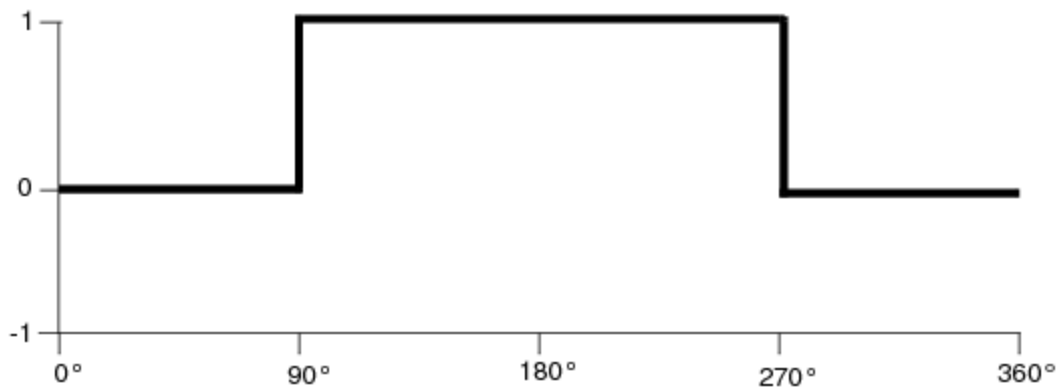
Pulse Primitive Duty Cycle Set to 20%

The following figure shows the effect of inverting a primitive while the duty cycle is set to 20. In this case, the primitive remains low for 20% of the component's period and the remaining 80% at the higher level.

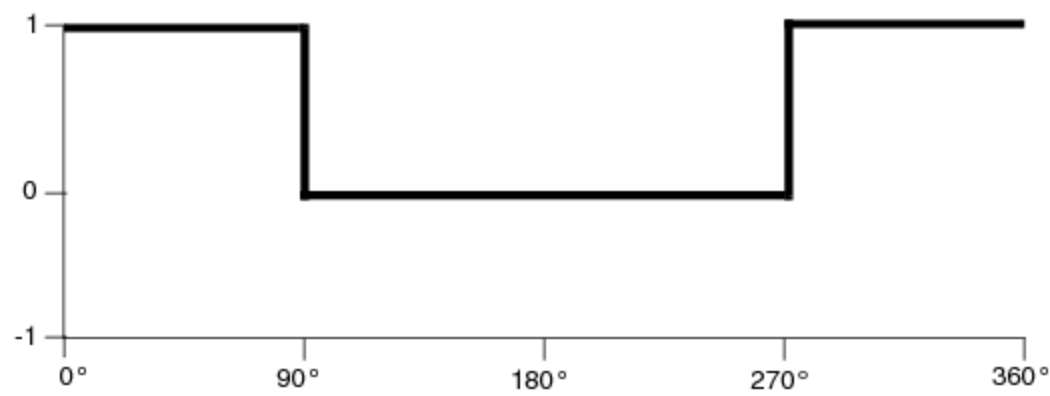


Pulse Primitive Duty Cycle Set to 20% While Inverted Phase

The phase of a pulse primitive refers to the delay before the first state of the pulse is executed. The phase is measured in degrees over one cycle (360 degrees) of the pulse primitive. The figure shows the phase delay that results from setting the phase to 90 degrees. The phase parameter's default value is 0.0 degrees.



Pulse Primitive Phase Set to 90 Degrees



Pulse Primitive Phase Set to 90 Degrees and Inverted

Invert pulse?

This parameter flips the orientation of a pulse primitive from high to low.

Yes	Inverts the signal.
No	Leaves the signal in its normal state. Default.

Maximum risetime (% of period)

The value specified in the Maximum risetime parameter is used by the Mixed Signal Workshop (MSW) to ensure that the MSW only offers timing solutions whose Sample Count (N) meets the primitive's rise time requirement. The relationship between the rise time and N is that the smaller the rise time limit, the larger the N must be to meet this limit. For example, a rise time of one percent requires at least 100 samples to meet this requirement.

Note that this parameter does not have an immediate effect on a Pulse primitive in WaveDesigner. Editing the value of this parameter does not change the plotted representation of this primitive in WaveScope.

Specify a value in the Maximum risetime parameter when you require a particular rise time: for example, if a pulse's rise time cannot be more than a particular percentage of the period. The maximum rise time value must be between 0.001% to 25% of the entire period. The default setting is 1%.

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Creating Wave Definitions : Using a Ramp Primitive

Using a Ramp Primitive

A Ramp primitive creates ascending or descending sloped waveforms between one and zero over the primitive’s entire period.

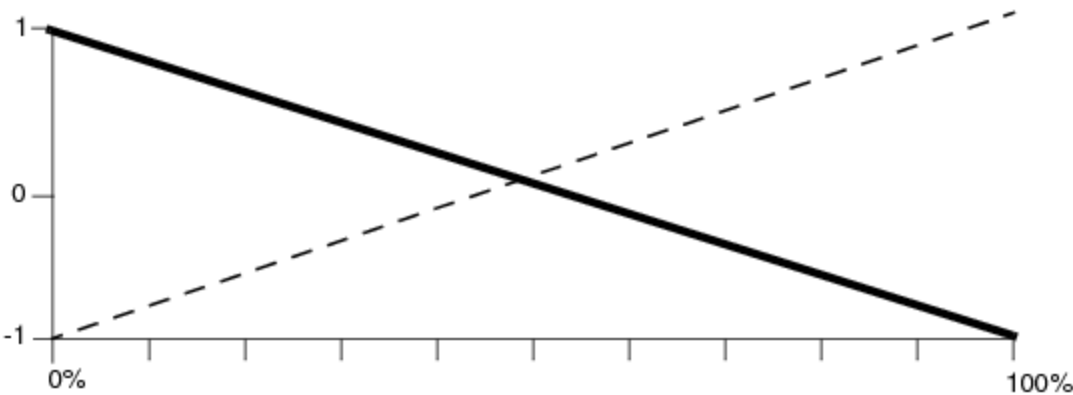
A Ramp primitive has two parameters that allow you to manipulate the positioning of the ramp primitive component.

- Invert ramp?
- Maximum risetime (% of period)

Invert ramp?

This parameter determines whether the slope of the ramp is positive or negative:

Yes	Signal is inverted or has a negative slope.
No	Signal is in its normal state as a positive slope. Default.



Inverting a Ramp Primitive

Maximum risetime (% of period)

The value specified in the Maximum risetime parameter is used by the Mixed Signal Workshop (MSW) to ensure that the MSW only offers timing solutions whose Sample Count (N) meets the primitive's rise time requirement. The relationship between the rise time and N is that the smaller the rise time limit, the larger the N must be to meet this limit. For example, a rise time of one percent requires at least 100 samples to meet this requirement.

Note that this parameter does not have an immediate effect on a Ramp primitive in WaveDesigner. Editing the value of this parameter does not change the plotted representation of this primitive in WaveScope.

Specify a value in the Maximum risetime parameter when you require a particular period for the ramp to become a stable rise. Therefore, the smaller the rise time value, the earlier the ramp becomes stable. The maximum rise time value must be between 0.001% to 25% of the entire period. The default setting is 1%.

		
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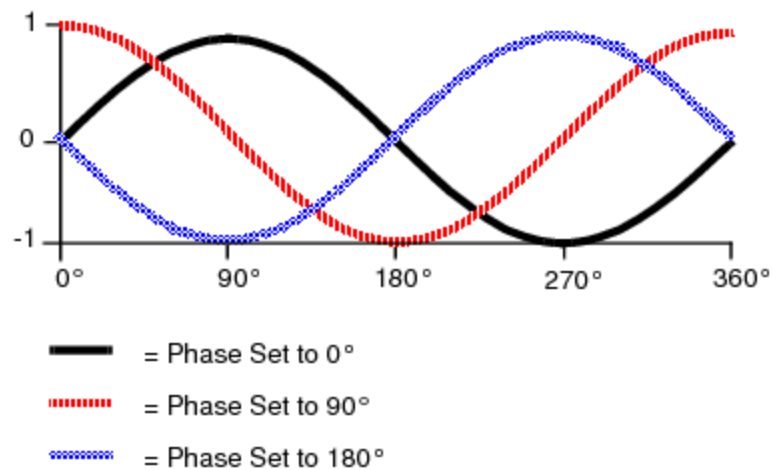


Creating Wave Definitions : Using a Sine Primitive

Using a Sine Primitive

A Sine primitive creates repeatable sinusoidal waveforms of arbitrary starting phase.

Phase is the only programmable parameter for the Sine parameter. This parameter allows you to control the relative starting position along the typical form of a sine wave. As shown in the figure, phase delay is measured in degrees, where 0 degrees starts at 0V. A 90 degree phase shift results in a Cosine starting at 1V.



Sine Primitive Phase Set to 0, 90, and 180 Degrees



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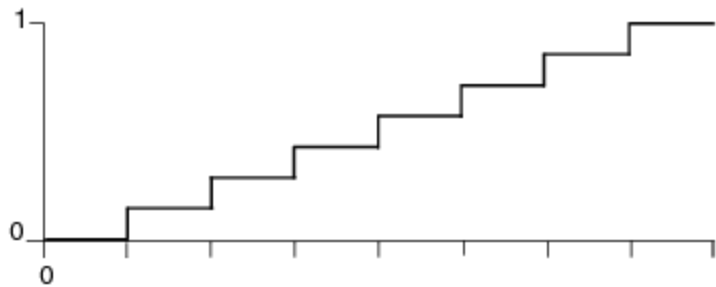
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Creating Wave Definitions : Using a Staircase Primitive

Using a Staircase Primitive

A Staircase primitive creates a ramp with discrete steps between 0 and 1.



Staircase Primitive

A Staircase primitive has these parameters:

Levels	The number of discrete steps in the staircase. The default is eight steps (levels).
Invert staircase?	Determines whether the slope of the staircase is positive or negative. - Yes: Signal is inverted or has a negative slope. - No: Signal is in its normal state as a positive slope. Default.
Maximum risetime as % of step time	This value is used by the Mixed Signal Workshop (MSW) to ensure that the MSW only offers timing solutions whose Sample Count (N) meets the primitive's rise time requirement. The relationship between the rise time and N is that the smaller the rise time limit, the larger the N must

be to meet this limit. For example, a rise time of one percent requires at least 100 samples to meet this requirement.

Note that this parameter does not have an immediate effect on a Staircase primitive in WaveDesigner. Editing the value of this parameter does not change the plotted representation of this primitive in WaveScope.

Specify a value in the Maximum risetime parameter when you require a particular rise time: for example, if the Staircase’s rise time cannot be more than a particular percentage of the step. The maximum rise time value must be between 0.001% to 25% of the step. The default setting is 1%.

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Creating Wave Definitions : Using a VBT Primitive

Using a VBT Primitive

A VBT primitive allows you to call a user-written VBT function that returns an array of samples. It allows the VBT function to become a custom primitive that accepts N, M, Fr, Fs, and other user-defined parameters.

A VBT primitive essentially gives you a handle to a DspWave object, which you then fill with samples based on the parameters in your VBT function. For example, to create a cosine, you can add the code in the example below to a module in the VBA Editor and set the VBT routine parameter in WaveDesigner to the name of the function (Cosine). The function then generates the shape of a cosine based on the VBT routine. See [Example VBT Primitive Function](#).

The following topics describe a VBT primitive’s component parameters and how to create a VBT routine for the VBT primitive:

- [VBT Primitive Component Parameters](#)
- [Creating a VBT Primitive](#)

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Creating Wave Definitions : Using a VBT Primitive : VBT Primitive Component Parameters

VBT Primitive Component Parameters

A VBT primitive generates an array of samples by calling a user-written VBT routine. The parameters are:

VBT routine	The name of the user-written VBT routine.
Bandwidth limited	If checked indicates that the waveform has a finite bandwidth, for example, it consists of one or more sinusoids (tones). Mixed Signal Workshop (MSW) uses this parameter and the Required BW multiplier parameter when a waveform is instantiated on an instrument.
Required BW multiplier	MSW assumes that the waveform’s bandwidth is its repeat rate (FR) multiplied by this constant. For band-unlimited waveforms, the 10%-90% edge rise time as a percent of the period can be approximated as: $tr \cong 0.35 / \text{RequiredBandwidthMultiplier}$
Fractional N capable	Select if the VBT routine can handle fractional values of N. This could occur if, for example the VBT primitive was added to a WaveDef containing a Sine. If 35 samples are generated, the VBT primitive will be asked to generate 17.5. If this is allowed by your routine, check this box. See the description of numSamples, N, and startingN in VBT Primitive Function Declarations.
Multiple Cycle Capable	Select if the VBT routine can handle multiple cycles.
Min samples	The minimum number of samples that the VBT routine can handle. Blank indicates that there is no minimum.

Max samples	The maximum number of samples that the VBT routine can handle. Blank indicates that there is no maximum. Set min = max to indicate a single value of N.
Volatile	If checked, the VBT routine will return different samples on each call. For example, a random number generator would be volatile.
Max Natural Envelope	If the maximum value generated by the VBT routine at amplitude 1, offset 0, is known, enter it in this field. This improves performance by preventing the software from having to search all samples.
Min Natural Envelope	If the minimum value generated by the VBT routine at amplitude 1, offset 0, is known, enter it in this field. This improves performance by preventing the software from having to search all samples.
Runtime Arg1 to 20	User defined arguments that are passed to the VBT procedure at run time.

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Creating Wave Definitions : Using a VBT Primitive : Creating a VBT Primitive

Creating a VBT Primitive

The following procedure creates a VBT primitive:

1. Add a module or choose a place in an existing module in VBA Editor.
2. Write a routine making sure that it has the proper signature format and content.
3. Open WaveDesigner.
4. Add a new wave definition with a VBT primitive.
5. Type the VBT routine’s name in the VBT routine box. Note that this is the VBT routine name, not the module name.
6. Set the parameters in the component parameter fields. For example, in the cosine example, check Bandwidth limited, Fractional N capable, and Multiple Cycle capable. The minimum and maximum natural envelope for the cosine is -1 and 1, respectively. To specify phase, put it in the Runtime Arg1 field.
7. Click the Show Wave button.

VBT Primitive Function Declarations

Declare a VBT primitive function as follows:

```
Function UserVBT(ByVal inWave As DSPWave, ByVal numsamples As Long, _
ByVal N As Double, ByVal startingN As Double, ByVal M As Long, _
ByVal Amplitude As Double, ByVal DCOffset As Double, _
ByVal Ft As Double, ByVal Fs As Double, _
ParamArray userArgs()) As String
End Function
```

The VBT routine parameters are:

inWave	An empty DspWave is created by IG-XL. You are given a handle to this DspWave object through inWave. Fill inWave with the samples to be sourced. DSPWave.
numSamples	The number of samples to be generated. Long.
N	If the Fractional N capable box was not selected, then N = numSamples. Double.
startingN	If combined with another primitive and passed a fractional N, this will be the fractional offset. Double.
M	The number of repetitions (cycles) of the signal to be generated. Long.
Amplitude	Signal amplitude. Double.
DCOffset	Signal DC Offset. Double.
Ft	Repetition rate of one cycle of the waveform in Hertz. Double.
Fs	Instrument sampling frequency. Double.
userArgs()	An array of up to 20 Strings containing the optional Args. Since the number of optional Args may change in the future, the routine should determine the actual number of array elements using the UBound function. String.

For example, if a wave definition is composed of a Sine primitive followed by a VBT primitive and asked to generate 35 samples, the Sine primitive generates 18 samples. In this case:

- NumSamples would be 17. This is the actual number of samples the VBT primitive will return in inWave.
- N would be 17.5, which is theoretically what the VBT was asked to generate (35/2), and this value should be used to help calculate the samples to return.

- `startingN` would be 0.5, as this is the amount of samples the Sine primitive has generated over its N (18-17.5).

Handling Errors

A well-written VBT primitive should check its arguments and signal an error if appropriate. Errors are signaled as follows:

1. Do not put any samples into the returned `DspWave`. The caller assumes an error has occurred based on the `SampleSize` of 0.
2. Set the function to a string that describes the error.

Example VBT Primitive Function

The following is an example of a VBT function that generates a cosine function. The starting phase delay (optional) is in Runtime Arg1.

```
Function Cosine(ByVal inWave As DspWave, ByVal numSamples As Long, _
    ByVal N As Double, ByVal startingN As Double, ByVal M As Long, _
    ByVal Amplitude As Double, ByVal DCOffset As Double, _
    ByVal Ft As Double, ByVal Fs As Double, _
    ParamArray userArgs()) As String
    ' Return cosine samples in the DspWave
    Dim i As Long, phaseString As String, phase As Double
    Dim smpls() As Double
    Static PI As Double
    If PI = 0 Then PI = 4 * Atn(1)
    ' Check that the starting phase is between 0 and 360 degrees
    phaseString = Trim$(userArgs(0))
    If phaseString = "" Then
        phase = 0#           ' Default is 0 degrees
    Else
        ' Check for errors
        If Not IsNumeric(phaseString) Then
            Cosine = phaseString & " is not a valid number"
            Exit Function
        Else
            phase = CDBl(phaseString)    ' Convert to Double
            If phase < 0 Or phase > 360 Then
```

```
Cosine = phaseString & " is not between 0 and 360 degrees"
```

```
Exit Function
```

```
End If
```

```
End If
```

```
End If
```

```
' Convert valid phase to radians
```

```
phase = PI * phase / 180
```

```
' Create a samples array
```

```
ReDim smpls(numSamples - 1)
```

```
' Fill it with samples
```

```
For i = 0 To numSamples - 1
```

```
    smpls(i) = Amplitude * Cos(2 * PI * i * M / numSamples + phase) _  
        + DCOffset
```

```
Next
```

```
' Return it in the DSPWave
```

```
inWave.Data = smpls
```

```
End Function
```

Example VBT Function Using DspWave FileImport Method

The following example reads data from a file using the `DspWave FileImport` method. It then normalizes the data and returns it in the `inWave` parameter. Note that the `inWave` parameter is passed in `ByVal`. This means that you are given a copy of the address of `inWave`. To perform DSP methods, use the `Data` property of the `DspWave` rather than using `Set` statements. As shown below, the data is normalized using:

```
inWave.Data = inWave.DivideScalar(inWave.CalcMaximumMagnitude).Data
```

If `Set inWave = inWave.DivideScalar(inWave.CalcMaximumMagnitude)` is used, you will lose the handle to the original `inWave` object and your changes will be lost.

```
Function FileImportwNormalization(ByVal inWave As DSPWave, _
```

```
    ByVal NumSamples As Long, ByVal N As Double, _
```

```
    ByVal startingN As Double, ByVal M As Long, _
```

```
    ByVal Amplitude As Double, ByVal DCOffset As Double, _
```

```
    ByVal Ft As Double, ByVal FS As Double, _
```

```
    ParamArray userArgs() As String
```

```
    Dim FileName As String
```

```
    Dim FileType As FileType
```

```
    Dim FileSampleRate As Double
```

```
' File name, type, and Sample Rate are passed in as user arguments to
```

```
' the primitive
```

```

FileName = userArgs(0)
FileType = userArgs(1)
FileSampleRate = userArgs(2)
' Import the data into inWave
inWave.FileImport FileName, FileType
inWave.Info.StorageLocation = "CPU"
' Convert the samples to doubles before normalization
inWave.Data = inWave.ConvertToReal.Data
inWave.SampleRate = FileSampleRate
' Normalize the waveform to +/-1 and apply amplitude and offset
inWave.Data = inWave.DivideScalar(inWave.CalcMaximumMagnitude / _
    Amplitude).AddScalar(DCOffset).Data

```

End Function

Passing Multiple User Arguments in a VBT Primitive Parameter

The usual procedure for passing user arguments in a VBT primitive parameter is to use one or more of the 20 userArgs() available in the function. Normally, you place one value in each userArgs parameter.

You can also pass more than one value in a userArgs by specifying multiple values separated by delimiters. The procedure is as follows:

1. On a Wave Definitions sheet, define a wave.
2. in the Primitive Parameters column, right click and select Edit to open the Primitive Parameters window.
3. Enter a delimited list of values in one of the Runtime Arg fields. For example, you could enter the following in Runtime Arg1: 24|25|26
4. In the VBT primitive function you break down the argument using syntax similar to the following:

```

Function waveform( ParamArray user_Args,... )
Dim FT_Args
Dim FT_Args1
FT_Args = Split(user_Args1, "|")
FT_Args1 = FT_Args(0)
End Function

```

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Creating Wave Definitions : Adding a Wave Definition at Run Time

Adding a Wave Definition at Run Time

Typically, you use WaveDesigner to create wave definitions when creating a test program. You may, however, need to create wave definitions at run time. In such a case, use the `.CreateWaveDefinition` VBT method to add a named wave definition to the WaveDefinitions collection at run time. You can then use this wave definition as the WaveDef Name in a Source instrument's Signal.

CreateWaveDefinition Method

The syntax for the WaveDefinitions method is:

`TheExec.WaveDefinitions.CreateWaveDefinition(WaveDefinitionName, samples, Volatile)`

This method adds a named wave definition to the WaveDefinitions collection. This method does not actually create an arbitrary wave definition. It creates wave definitions that are really hardcoded samples.

Use an array of doubles, a `DspWave`, or an `InstrWave` to create samples. Then call this method with a unique `WaveDefinitionName`. Multiple calls to this method using the same `WaveDefinitionName` overwrite the samples. However, you must set the `Volatile` parameter to `True` to overwrite samples. An error is returned if you use a `WaveDefinitionName` that duplicates a name used on a Wave Definitions sheet.

An error also occurs if you call this method twice using the same `WaveDefinitionName`, without setting the `Volatile` parameter to `True`.

For more information, see `TheExec.WaveDefinitions.CreateWaveDefinition` in the VBT Syntax Reference.

Using the CreateWaveDefinition Method at Run Time

The following procedure describes how to use the `.CreateWaveDefinition` method to add a wave definition to the Waveform collections at run time:

1. [Create a subroutine that creates samples and retrieves them from a file. Within this routine call .CreateWaveDefinition to create the named wave definition.](#)

For example:

```
Sub CreateWaveDefFromFile()  
    ' Using the DSPWave's FileImport method  
    ' First Create a new DSPWave  
    Dim inWave As New DSPWave  
    ' Import an IMAGE wav file  
    inWave.FileImport ".\bbacsrcSeg_01.wav", File_Image_wave  
    inWave.Info.StorageLocation = "CPU"  
    ' Convert the samples to doubles before normalization  
    inWave = inWave.ConvertToReal  
    inWave.SampleRate = 8832000  
    ' Normalize the waveform to +/-1  
    inWave = inWave.DivideScalar(inWave.CalcMaximumMagnitude)  
    ' Call CreateWaveDefinition to add the named wave definition  
    ' to the WaveDef collection  
    TheExec.WaveDefinitions.CreateWaveDefinition "myCustomWaveDef", inWave  
End Sub
```

2. [Decide when you want this routine called. If you want this wave definition to be used across several jobs, call this routine in an executive interpose function, such as OnProgramValidated. If you want the samples to be recreated for each run, mark the wave definition as volatile by setting the Volatile flag of the CreateWaveDefinition method to True.](#)

[You can use this wave definition name in the same manner as those created on a Wave Definitions sheet. For example, you can use it in a Source instrument’s Signal.](#)

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WaveDesigner Display

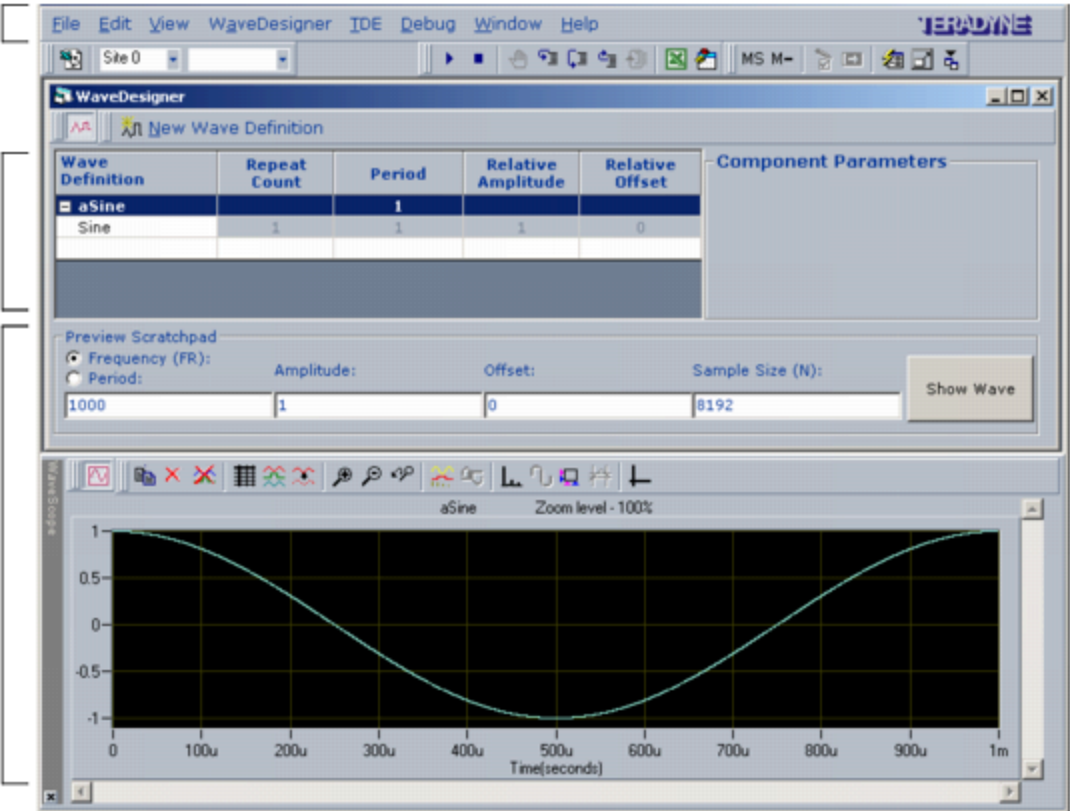
WaveDesigner Display

WaveDesigner is a tool you use to create simple or compound wave definitions which define a waveform's shape. Creating and editing the characteristics of a wave definition involves adding primitives to wave definitions, defining component parameters, and fine tuning each primitive component relative to one another.

Command Menu

Wave Definition Table and Component Parameters

Scratchpad & WaveScope



WaveDesigner Display

To open WaveDesigner from the Test Debug Environment (TDE), select TDE>Tools>WaveDesigner. The following topics are available:

•	WaveDesigner Menu Commands
•	Wave Definition Table
•	Component Parameters
•	Preview Scratchpad and WaveScope

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WaveDesigner Display : WaveDesigner Menu Commands

WaveDesigner Menu Commands

The WaveDesigner menus include commands to create, edit, and apply wave definitions. The following topics are available:

- | | |
|---|-----------------------------------|
| • | File Menu |
| • | Edit Menu |
| • | View Menu |
| • | WaveDesigner Menu |

Note:
Since the WaveDesigner window includes both the WaveDesigner tool and the WaveScope tool, the window’s menu bar toggles between the two tools depending upon which is active. To toggle between the two menu bars, click anywhere within the tool that you want active. See [WaveScope Toolbar and Command Reference](#) for information about WaveScope commands.

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WaveDesigner Display : WaveDesigner Menu Commands : File Menu

File Menu

File menu commands allow you to open a new wave definition, save active wave definitions, and close the WaveDesigner or Test Development Environment (TDE).

Close WaveDesigner		Closes the active WaveDesigner display.
New Wave Definition		Appends a new wave definition to the Wave Definition table. See Inserting a New Wave Definition.
Save to Excel		Saves the wave definition in the Wave Definition table to a Wave Definitions sheet in the IG-XL workbook.
Exit TDE		Closes all active TDE tools and displays in one action.

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WaveDesigner Display : [WaveDesigner Menu Commands](#) : Edit Menu

Edit Menu

Edit menu commands allow you to add, remove, order, or rename primitives and wave definitions in the Wave Definition table.

Insert>Primitive

Inserts a primitive in the selected wave definition above the selected line.

The Insert command is only active when a primitive line in the Wave Definition table is selected.

- **DC**: Inserts a DC primitive into the selected wave definition.
- **ExcelRange**: Inserts an ExcelRange primitive into the selected wave definition. Its parameters call a named range in the workbook to create waveform samples.
- **MultitoneLab**: Starts the MultiTone Lab editor which is used to create a simple multitone waveform.
- **Pulse**: Inserts a Pulse primitive which generates high or low pulses.
- **Ramp**: Inserts a Ramp primitive which creates ascending or descending slope waveforms.

- **Sine**: Inserts a Sine primitive into the selected wave definition to create a sinusoidal waveforms of arbitrary starting phase.
- **Staircase**: Inserts a Staircase primitive which creates ramps with discrete steps.
- **VBT**: Inserts a VBT primitive whose parameters invoke a VBT routine that returns an array of samples.

Insert>Wave Definition	Inserts an existing wave definition in the selected wave definition to create a compound wave definition. This command is only active when a primitive line in the Wave Definition Table is selected.
Delete	(Ctrl+D) Removes the selected wave definition from the wave definition table or the selected primitive.
Rename	(F2) Edits the name assigned to the selected wave definition.
Move Down	This functionality has not been implemented.
Move Up	This functionality has not been implemented.

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WaveDesigner Display : WaveDesigner Menu Commands : View Menu

View Menu

View menu commands allow you to view the waveforms created by the wave definitions in the WaveScope display at the bottom of the WaveDesigner window. Note that when you select any command from the View menu, the WaveDesigner menu toggles over to the WaveScope menu providing control over the WaveScope tool. To switch back to WaveDesigner, click anywhere in the WaveDesigner region of the window.

Show Wave	Displays a waveform in the WaveScope display based on the parameters of the selected wave definition.
Show All	Displays all waveforms listed in the Wave Definition table in succession in the WaveScope. You must activate Multi-Channel mode in WaveScope to view all waveforms.
Clear Scope	Removes all waveforms displayed in the WaveScope display.



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WaveDesigner Display : WaveDesigner Menu Commands : WaveDesigner Menu

WaveDesigner Menu

WaveDesigner menu commands toggle between Specify Total Period and Calculate Total Period.

Specify Total Period	Allows you to determine the total period of the wave definition. See Specifying the Period.
Calculate Total Period	Calculates the sum of the specified period for each primitive in the wave definition. See Wave Definition’s Total Period.

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WaveDesigner Display : Wave Definition Table

Wave Definition Table

The Wave Definition table is the area of the display where you build wave definitions. Within this table structure, you create a wave definition and populated it with primitives and nested wave definitions to produce the shape identity of a waveform. Within a wave definition, the repeat count, period, amplitude, and offset are represented as being relative to one another.

The wave definition table includes the following column headings:

Wave Definition	Either contains the wave definition name or the names of its founding components (primitives and nested wave definitions)
Repeat Count	Defines the number of times that the primitive is sequentially created. See Specifying the Repeat Count.
Period	The period of the component relative to the total period of the wave definition. See Specifying the Period.
Relative Amplitude	Amplitude of each component relative to the first component. See Specifying the Relative Amplitude.
Relative Offset	The offset of this component scaled by the relative amplitude of the first component. See Resulting Compound Wave Definition Using the Relative Amplitude.

The following sections provide instructions for accomplishing various tasks using the wave definition table.

- Inserting and deleting primitives:

– [Inserting a New Wave Definition](#)

– [Inserting a Primitive](#)

– [Creating a Compound Wave Definition](#)

– [Deleting a Primitive or a Nested Wave Definition](#)

– [Renaming a Wave Definition](#)

• [Specifying wave definition table parameters:](#)

– [Specifying the Repeat Count](#)

– [Specifying the Period](#)

– [Wave Definition's Total Period](#)

– [Specifying the Relative Amplitude](#)

– [Resulting Compound Wave Definition Using the Relative Amplitude](#)

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WaveDesigner Display : Wave Definition Table : Inserting a New Wave Definition

Inserting a New Wave Definition

To create a new wave definition, you insert a new row in the Wave Definition table. The initial row with an expand/collapse button represents the wave definition row in the table.

To insert a new wave definition:

1. Open the New Wave Definition dialog box by doing either of the following:
- Click the New Wave Definition button in the toolbar.

- Select File>New Wave Definition.

The New Wave Definition dialog box appears.

2. Type a Wave Definition Name and select a primitive from the Primitive menu.

Note that selecting the empty field avoids inserting a primitive and creates an empty wave Definition.

Waveform names are case-insensitive when you create them, but references to them in VBT must match the case of the name given here. See Case Sensitivity of Wave Definition Names in the VBT Syntax Reference.

3. Click OK.

An alternative method for inserting a new wave definition is:

1.

Right-click a row in the Wave Definition Table.
2.

From the context menu that now appears, select Create New Wave Definition.
3.

Complete steps 2 and 3 as described above.

Note that the new wave definition is added at the end of the current wave definition table.

☒ **Note:**
The new wave definition does not appear on a Wave Definitions sheet until you save it using the File>Save command.

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WaveDesigner Display : Wave Definition Table : Inserting a Primitive

Inserting a Primitive

You can create wave definitions that consist of multiple primitive components by inserting additional primitives to the wave definition group in the Wave Definition table.

To insert additional primitives in an existing wave definition, do either of the following:

- Click below the cell where you want to place the new primitive. Select Edit>Insert>Primitive and then select a primitive type.
- Right-click the below the cell where you want to place the new primitive. From the context menu that now appears, select Insert>Primitive, and select a primitive type.

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WaveDesigner Display : Wave Definition Table : Creating a Compound Wave Definition

Creating a Compound Wave Definition

You can embed an existing wave definition in another wave definition to create a compound wave definition. A compound wave definition can consist of one or more embedded wave definitions. To create a compound wave definition:

1. Insert a new wave definition. See [Inserting a New Wave Definition](#).
2. Use either of the following procedures to insert additional wave definitions in an existing wave definition:
 - Click below the cell where you want to insert the wave definition. Select [Edit>Insert>Wave Definition](#) and then select the wave definition name.
 - Right-click below the cell where you want to insert the wave definition. From the context menu that now appears, select [Insert>Wave Definition](#), and select the wave definition name.

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WaveDesigner Display : Wave Definition Table : Deleting a Primitive or a Nested Wave Definition

Deleting a Primitive or a Nested Wave Definition

To delete a primitive or a nested wave definition from within a wave definition:

- | | |
|----|--|
| 1. | Select the cell of the primitive or the nested wave definition to be deleted from the wave definition table. |
| 2. | Select Edit>Delete. |

☒ **Note:**

You cannot delete a primitive or a wave definition from the wave definition table by highlighting the row and pressing Delete. In fact, this action clears the WaveScope display.

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WaveDesigner Display : Wave Definition Table : Renaming a Wave Definition

Renaming a Wave Definition

To edit the existing name of a wave definition:

1.

Select the cell of a wave definition from the wave definition table.
2.

Select Edit>Rename or double-click the wave definition name.
3.

Type in the new name in the Wave Definition column and press Enter.

or:
Single-click the Wave Definition name in the wave definition table and type the new name. Click away from the cell or press Enter to complete the action.

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WaveDesigner Display : Wave Definition Table : Specifying the Repeat Count

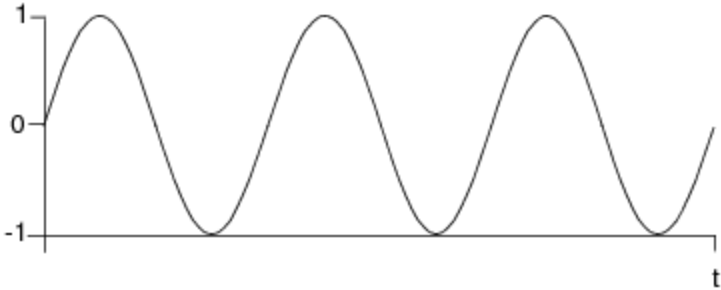
Specifying the Repeat Count

☒ **Note:**

Using Repeat Count is only recommended when creating compound wave definitions. For simple wave definitions, specify Repeat Count (M) on the Mixed Signal Timing sheet or using VBT syntax. This helps to avoid duplicate declarations of the Repeat Count value.

The Repeat Count parameter defines the number of times that a primitive is sequentially created. Specifying a repeat count prevents you from having to insert two or more identical primitives when you want consecutive identical primitives.

The figure shows a single sine primitive with a repeat count. Note that the same waveform could have been created by inserting three consecutive sine primitive components. However, specifying the primitives repeat count as 3 is more efficient and gives the same result.



Sine Primitive Component with a Repeat Count of 3

To specify the Repeat Count for an existing primitive:

1. Select the Repeat Count column cell that corresponds to the primitive to be edited.

2. Type the number of times that you want the primitive to repeat consecutively.

3.

Click Enter.
4.

Click the WaveScope's Show Wave button to view the results of editing the primitive's Repeat Count.

Note:
Wave definition primitives are relative to one another over the length of the wave definition's period. Increasing the Repeat Count parameter of a primitive component only increases the number of times that the primitive appears within that component. It has no effect on the relative size of the primitive component or the wave definition. Use the Period parameter to adjust the primitive's relative length within the wave definition. See [Specifying the Period](#) for more information. Also, Repeat Count does not change the number of samples used to generate the wave definition; each cycle will now receive N/M samples.

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WaveDesigner Display : Wave Definition Table : Specifying the Period

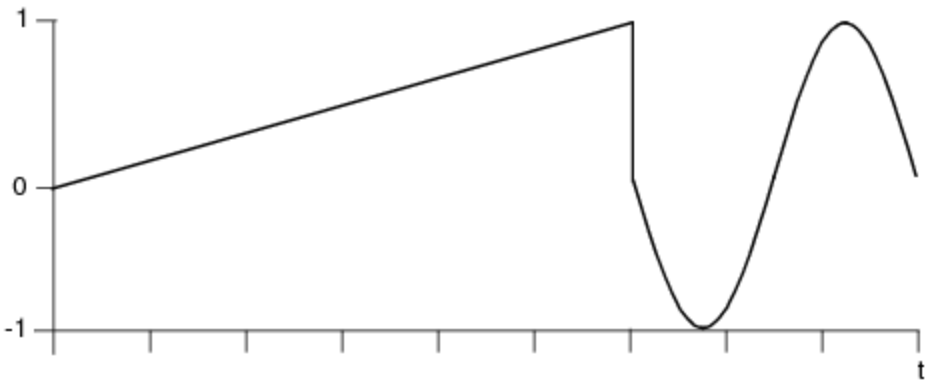
Specifying the Period

☒ **Note:**

The Period column is only used when creating compound wave definitions. It is irrelevant for simple wave definitions.

When building a wave definition, you can control the relative length of each primitive that makes up the total waveform. A primitive's Period value in the Wave Definition table indicates how long that component is relative to the others in that wave definition. The primitive value represents a ratio portion of the entire length of the wave definition.

For example, the figure shows a wave definition that consists of a ramp primitive and a sine primitive. The ramp component has a period of 2 and sine has a period of 1. In this case, the ramp component of the total waveform is twice as long as the sine wave.



Ramp Primitive with a Period Value of 2

To specify the Period of an existing primitive component:

1. Select the Period column cell that corresponds to the primitive to be edited.

2.

Type the number that represents the primitive’s relative length compared to the total period of the wave definition.
3.

Click Enter.
4.

Click the WaveScope’s Show Wave button to view the results of editing the primitive’s Period.

Note:

The sum of each primitive’s period is automatically calculated to become the wave definition’s total period. However, you can also control the value of the wave definition’s total Period by using the Specify Total Period command. See [Wave Definition’s Total Period](#) for more information.

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WaveDesigner Display : Wave Definition Table : Wave Definition’s Total Period

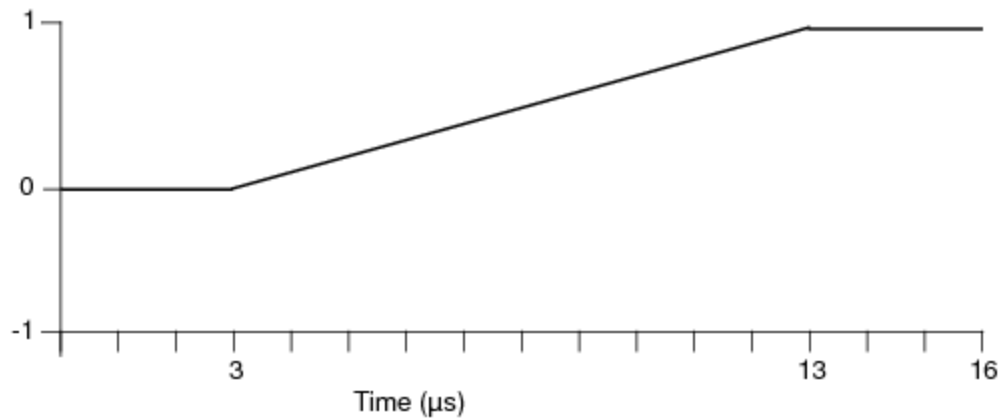
Wave Definition’s Total Period

The length of each primitive component in a waveform definition is relative to the total length of the wave definition’s entire period. By default, the length of the wave definition always remains the number of primitives and wave definitions that make up the wave definition. The Total Period value of the waveform represents sum of the period values for each of the primitives. For example, the table and figure show a situation where a compound wave definition ratio requirements are:

- 3 units of a DC primitive
- 10 units of a Ramp primitive
- 3 units of additional DC primitive

Period Parameter Entries for a Compound Wave Definition

Wave Definition	Repeat Count	Period	Relative Amplitude	Relative Offset
myWave		16		
DC	1	3	1	-1
Ramp	1	10	1	0
DC	1	3	1	0



Resulting Compound Wave Definition Using the Period Parameter

Setting the Wave Definition table parameters to the values in the table and then clicking the Show Wave button displays the intended waveform, but the scale of the WaveScope's x-axis is not scaled to demonstrate microseconds. Therefore, you must select Period in the Preview Scratch pad and type 0.000016 in the Period box. Click the Show Wave button again, and the intended wave definition is plotted in its correct format. Note that this value is not saved, it is only for previewing the waveform.

Specifying and Calculating Total Period

The Wave Definition table automatically calculates a wave definition's Total Period. However, WaveDesigner also provides a toggle command that allows you to specify a particular Total Period or to calculate the Total Period based on the existing primitive period values.

The Specify Total Period command allows you to predetermine the total period of a wave definition. You might specify the total period if you want to build the wave definition one primitive at a time. Creating wave definitions in this way ensures that the wave definition primitives never exceed the desired total period.

By toggling Specify Total Period, you access the Calculate Total Period command which performs an automatic summation of the primitive periods within the wave definition and updates the Wave Definition table's Period cell for the wave definition.

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WaveDesigner Display : Wave Definition Table : Specifying the Relative Amplitude

Specifying the Relative Amplitude

☒ **Note:**

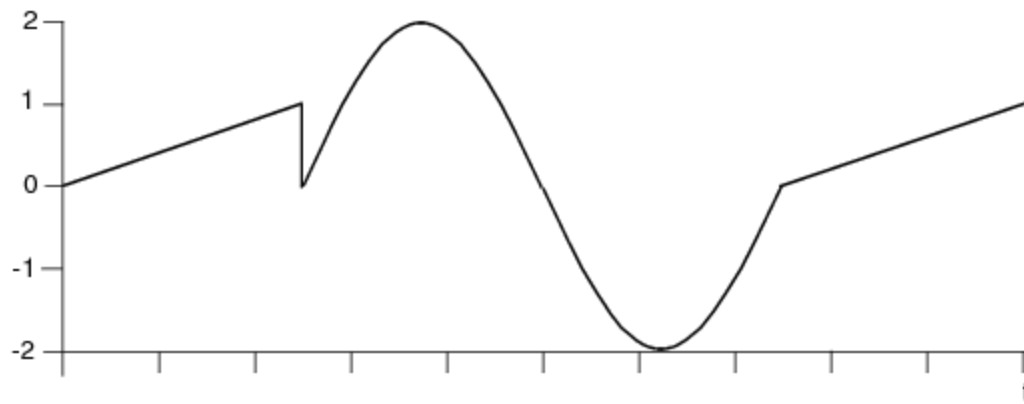
The Relative Amplitude column is only used when creating compound wave definitions. It is irrelevant for simple wave definitions.

The Wave Definition table includes a Relative Amplitude column. The relative amplitude parameter means that the amplitude of each primitive component is relative to the first component in that wave definition.

For example, the table and figure show that the amplitude of the Sine wave is twice that of the first Ramp component. The amplitude of the last ramp is equal to the first since the relative amplitude value is the same.

Relative Amplitude Parameter Entries for a Compound Wave Definition

Wave Definition	Repeat Count	Period	Relative Amplitude	Relative Offset
myWave		4		
Ramp	1	1	1	0
Sine	1	2	2	0
Ramp	1	1	1	0



Resulting Compound Wave Definition Using the Relative Amplitude

To specify the Relative Amplitude of an existing primitive component:

1. [Verify the Relative Amplitude value for the first component of the wave definition.](#)
2. [Select the](#) Relative Amplitude column cell that corresponds to the primitive to be edited.
3. [Type the number that represents the primitive's relative amplitude compared to the relative amplitude of the first component of the wave definition.](#)
4. [Click](#) Enter.
5. [Click the WaveScope's](#) Show Wave button to view the results of editing the primitive's Relative Amplitude.

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WaveDesigner Display : Wave Definition Table : Specifying the Relative Offset

Specifying the Relative Offset

☒ **Note:**

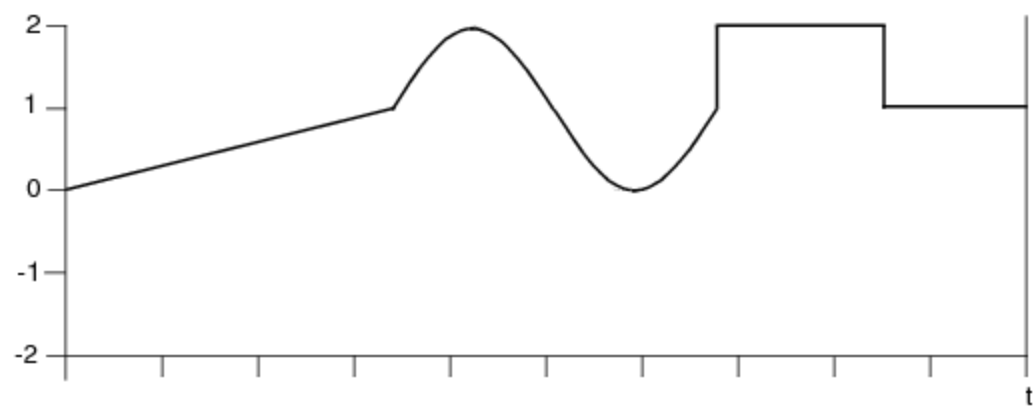
Use Relative Offset only when creating compound wave definitions. For simple wave definitions, specify your Offset on the Signal test element. This helps to avoid specifying your offset in more than one place.

The Wave Definition table includes a Relative Offset column. The relative offset parameter means that the offset of each primitive component is relative to the amplitude of the first component in that wave definition.

The table and figure show an example of how the relative offset is applied to primitives. The offset of the sine primitive is calculated by dividing the amplitude from the Preview Scratchpad by the Relative Amplitude of the Ramp. The Sine's Relative Offset is then multiplied by this product. The final Offset is reached by adding the offset from the Preview Scratchpad to this number.

Relative Offset Parameter Entries for a Compound Wave Definition

Wave Definition	Repeat Count	Period	Relative Amplitude	Relative Offset
myWave		3		
Ramp	1	1	1	0
Sine	1	1	1	1
Pulse	1	1	1	1



Resulting Compound Wave Definition Using the Relative Offset
Using this formula, the primitives remain relative to each other.

For example, assume that the Amplitude in the Preview Scratchpad is 1, and the offset is 0. The Sine's offset is 1 multiplied by the Preview Scratchpad Amplitude over the relative amplitude of the Ramp, which is 1. When added to our Preview Scratchpad offset of 0, this results in a final offset for the Sine of 1. The Pulse primitive component has a relative offset value of 1 and an amplitude relationship to the first component of 1. That means that the Pulse's offset is 1.

To specify the Relative Offset of an existing primitive component:

- | | |
|----|---|
| 1. | Verify the Relative Offset value for the first component of the wave definition. |
| 2. | Select the Relative Offset column cell that corresponds to the primitive to be edited. |
| 3. | Type the number that represents the primitive's relative offset compared to the relative amplitude of the first component of the wave definition. |
| 4. | Click Enter. |
| 5. | Click the WaveScope's Show Wave button to view the results of editing the primitive's Relative Offset. |

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WaveDesigner Display : Component Parameters

Component Parameters

Component parameters are primitive-specific parameters that allow you to control aspects of a signal’s shape characteristics or to influence how WaveDesigner or the Mixed Signal Workshop tools treat the primitive.

The component parameters for the primitives are defined in [Creating Wave Definitions](#).

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WaveDesigner Display : Preview Scratchpad and WaveScope

Preview Scratchpad and WaveScope

Preview Scratch pad and WaveScope allow you to preview the waveform that you are creating with the wave definition. Since wave definitions are generic and do not have a determined frequency, amplitude, offset, or sample restrictions, the scratchpad allows you to associate temporary values to produce the waveform. You can then plot and analyze the waveform using the WaveScope display. The Preview Scratchpad consists of the following parameters and buttons:

Frequency (FR)	The frequency of the wave definition. Scales the x axis of the scope.
Period	The period of the wave definition: Scales the x axis of the scope.
Amplitude	The amplitude of the entire wave definition. The first component's relative amplitude is scaled to the actual amplitude, and the other components' amplitudes are scaled according to their relative amplitude relationship to the first component.
Offset	The offset from zero of the entire wave definition.
Samples Size (N)	The number of samples to generate the preview.
Show Wave	Displays the currently selected wave definition on the scope.

To display the waveform on the scope, click the Show Wave button or select View>Show Wave. See About WaveScope for instructions on using the WaveScope.

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